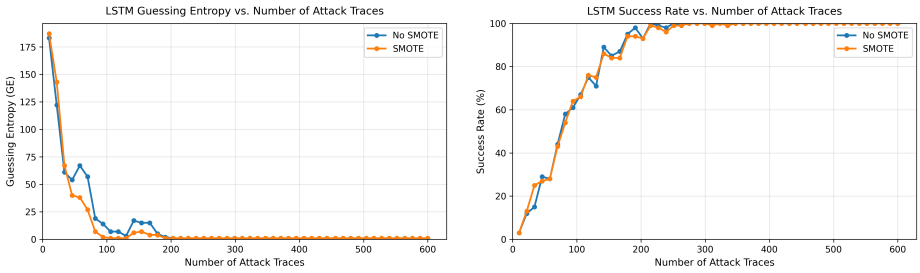
Auxiliary Public Dataset Baseline (AESPTv2) Evaluation

We use AESPTv2(`D1`)[1] as an off-board feasibility baseline. We attack byte 16 with 9 HW labels $c = \mathrm{HW}(S(x_{16} \oplus k_{16})$ where S denotes the first-round S-box) and use a fixed region-of-interest in leakage time samples ROI= [1150:1250], zero-centered. Compact CNN/LSTM classifiers output per-trace probabilities $p_i(c)$. [2][1] For each key guess k, we score $L(k;N) = \sum_{i=1}^{N} \log(p_i(c_i^{(k)}) + \varepsilon)$, where $c_i^{(k)} = \mathrm{HW}(S(x_{16}^{(i)} \oplus k))$ and $\varepsilon = 10^{-9}$. Ranking by $L(k;N)$ yields $\mathrm{GE}(N)$ and $\mathrm{SR}(N)$. This section validates the evaluation protocol.

We evaluate each CNN/LSTM model using the same region of interest (ROI) to ensure consistent results. For each key guess k we aggregate the sum of log-probabilities $L(k;N)$ across N traces and rank 256 candidates.

The full GE/SR curves are provided below: LSTM in Figs. 1a–1b and CNN in Figs. 2a–2b.



(a) Guessing Entropy vs. N (AESPTv2, byte 16, ROI [1150:1250]).

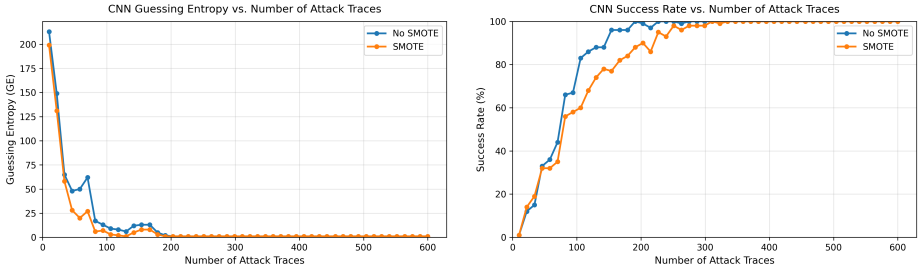
(b) Smoothed Success Rate vs. N (AESPTv2, byte 16, ROI [1150:1250]).

Fig. 1: LSTM evaluation on AESPTv2 for byte 16 over ROI [1150:1250]. The left plot reports Guessing Entropy (GE), and the right plot reports smoothed Success Rate (SR), comparing the SMOTE and no-SMOTE settings.

Table 1: Sample efficiency thresholds on AESPTv2 (approx. from Figs. 1a, 1b, 2a, 2b).

Model (variant)	$N_{\mathrm{GE}=1}$	$N_{\mathrm{SR}=90}$
LSTM (SMOTE)	∼100	∼180–200
LSTM (no SMOTE)	∼220	∼250–280
CNN (SMOTE)	∼220	∼240–260
CNN (no SMOTE)	∼210–230	∼200

[1] `https://github.com/urioja/AESPTv2`



(a) Guessing Entropy vs. N (AESPTv2, byte 16, ROI [1150:1250]).

(b) Smoothed Success Rate vs. N (AESPTv2, byte 16, ROI [1150:1250]).

Fig. 2: CNN evaluation on AESPTv2 for byte 16 over ROI [1150:1250]. The left plot reports Guessing Entropy (GE), and the right plot reports smoothed Success Rate (SR), comparing the SMOTE and no-SMOTE settings.

Baseline role of AESPTv2. The AESPTv2 experiments serve as a methodological baseline rather than as the paper's main cryptographic claim. They show that, with one leakage instance at a time, the GE/SR ranking protocol and compact neural models can already recover meaningful key rankings from a standard public benchmark. This establishes that the pipeline itself is sound. It also provides the right comparison point for the protocol-faithful ARM Cortex-M paired-leakage results in the accompanying paper: if single-instance profiling is already trace-hungry, then any sharp reduction in N under two-pair or eight-pair fusion can be interpreted as evidence that protocol-induced structure is being exploited rather than merely rediscovered.

References

1. L. Lerman, G. Bontempi, and O. Markowitch. A machine learning approach against a masked AES. *Journal of Cryptographic Engineering*, 5(2):123–139, 2015.
2. L. Masure, C. Dumas, and E. Prouff. A comprehensive study of deep learning for side-channel analysis. *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2020(1):348–375, 2020.